

17th Aug 2024

THOTA SRIRAM NAIDU
Vellore Institute of Technology
Vellore

Dear Thota Sriram Naidu,

Subject: Letter of Evaluation

This is with reference to the Global Academic Internship Programme (GAIP) conducted by Corporate Gurukul from 8th June 2024 to 29th June 2024 on 'Big Data Analytics using Deep Learning'. The course work for internship included the following:

- Introduction to Data Analytics
 - What is Data Analytics
 - Types of Data Analytics
 - Data in Data Analytics + Decision Models – Data Mining Process
 - Exploratory Data Analysis
 - Data Visualization
 - Data Querying
 - Statistical Methods for Summarizing Data
 - Exploring data using Pivot Tables
- Descriptive Statistical Measures
 - What is Descriptive Analytics?
 - Populations and Samples
 - Measures of Location
 - Measures of Dispersion
 - Measures of Shape
 - Measures of Association
- Regression
 - Introduction to Regression Analysis
 - Simple Linear Regression
 - Multi Linear Regression
 - Stepwise Regression
 - Coding Scheme for Categorical Variables
 - Problems with Linear Regression
- Introduction to Classification
 - Classification
 - Decision Trees
 - Bayesian Classifier
 - Logistic Regression
 - Support Vector Machine
 - Machine Separating
 - Hyperplane
 - Maximal Margin Classifier
 - Support Vector Classifier
 - Resampling Methods

- Introduction to Clustering
 - Affinity Measures and Partition Methods
 - K-means
 - K-medoids
 - Hierarchical Methods
 - Introduction to Association
 - Structure and Representation of Association Rules
 - Strong Association Rules and the Concept of Frequent Item sets – Apriori Algorithm
 - FP Growth
 - Time Series Analysis
- Introduction to Text Mining
 - Text Mining Terminologies
 - Text Mining Concepts
 - Text Mining Process
 - Knowledge Extraction Methods for Text Mining
 - Classification
 - Clustering
 - Association
- Artificial Neural Networks (ANN)
 - Introduction
 - Perceptron's and Activation Functions
 - Building Blocks of Neural Networks (Input, Hidden, Output layers)
 - Hands-On Exercise: Building a Simple Neural Network (XOR Problem)
 - How to develop ANN using TensorFlow
 - Back-propagation
 - Gradient Descent, Momentum, Learning Rate, Overfitting
- Convolutional Neural Networks (CNN)
 - Convolution, pooling operations
 - Popular CNN architectures
 - Applications of CNN in Python
- Recurrent Neural Networks (RNN)
 - Vanilla RNN
 - LSTM and GRU
 - Applications of RNN in Python
- Generative AI
 - Part 1: Introduction to Generative Models:
 - What are generative models?
 - Overview of model types: GANs, VAEs, and transformers.
 - Part 2: GANs (Generative Adversarial Networks)
 - Basic architecture: Generator and Discriminator.
 - Simple applications: Image generation, music generation (hands-on)
 - VAEs (Variational Autoencoders)
 - Encoder-decoder architecture.
 - Use cases: Image denoising.
 - Transformers in Generative AI
 - Overview of transformer architecture
 - Applications: Text generation with GPT models.

Part 3: Applications and Ethical Considerations

- Highlighting practical applications of each model type.
 - Brief discussion on ethical implications like deepfakes and bias.
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- Reinforcement Learning (RL)
 - Introduction
 - RL Algorithms
 - Dynamic Programming (DP)
 - Monte Carlo Methods
 - Temporal-Difference Learning (TD)
 - Implementing RL algorithm using Python

Your performance in GAIP was evaluated based on theoretical understanding and application of concepts in practical data analysis with **GRADE A**.

We encourage you to further your knowledge, skills and research in the above areas and wish you the very best for a career ahead!

Sincerely,



Dr Tan Wee Kek
Associate Professor
School of Computing
National University of Singapore



Dr Manoranjan Dash
Senior Data Scientist
School of Computing
National University of Singapore

TRANSCRIPT

GLOBAL ACADEMIC INTERNSHIP PROGRAMME – JUNE 2024 BIG DATA ANALYTICS USING DEEP LEARNING	
Name: THOTA SRIRAM NAIDU	Date: 17 th August 2024

Assessment Component	Score	Topic/Parameter
In-Class Assessment	12/40	Introduction to Data Analytics and Descriptive Statistical Measures
	24/40	Introduction to Regression Analysis and Classification
	24/40	Introduction to Clustering and Text Mining
	36/40	Artificial, Convolutional Neural Networks and Recurrent Neural Networks
Final Comprehensive Assessment	44/80	Comprehensive Assessment for the Course
Project Assessment	44/50	Final Project Work

	Assessment			Overall Percentage (Out of 100%)
	In-Class Assessment	Final Comprehensive Assessment	Project Assessment	
	30% weightage	20% weightage	50% weightage	73
Percentage	18/30	11/20	44/50	Grade A
Faculty Assessor Signature				-
Faculty Assessor Name	Dr Tan Wee Kek		Dr Manoranjan Dash	-

Grading Guideline:

O	100 - 90	B	54.9 - 50
A+	89.9 - 80	B-	49.9 - 45
A	79.9 - 70	C	44.9 - 40
A-	69.9 - 60	F	<40
B+	59.9 - 55		